Projective Temporal Ordering and Cumulative Projected Capacity: Numerical Evidence from the Spectral Admissibility Cascade

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Abstract

We identify a monotone proxy for temporal ordering within the Cosmochrony spectral admissibility programme. The natural observable along the BFS cascade is not the residual capacity $\sigma_{\text{pair}}(n)$, which decays by construction, but the cumulative projected capacity $I(n) = \sigma_{\text{pair}}(0) - \sigma_{\text{pair}}(n)$, which measures the spectral structure already projected at depth n. For $q \in \{29, 61, 101, 151\}$, $I(n)$ is found to be strictly non-decreasing at every BFS step, for every conjugate pair $(c, q-c)$, with zero physically meaningful regressions observed across 169 pairs and 310 steps after burn-in. This supports the identification of temporal ordering with the monotone accumulation of cumulative projected capacity along the admissible cascade. The saturation plateau of $I(n)$ is interpreted as Born–Infeld saturation at the pair level.

1 Motivation

The Cosmochrony framework identifies observable dynamics not with an evolution of the relational substrate χ—which is static—but with an iterated reprojection sequence $U_{t+1} = \Pi \circ \sigma(U_t)$, where Π is a fixed non-injective projection and $\sigma(U_t) \in \Pi^{-1}(U_t)$ selects an admissible configuration within the fiber compatible with the current effective state U_t [1]. This formulation raises a concrete question: what scalar observable, defined along the cascade, plays the role of a temporal ordering parameter? A candidate must be monotone along every admissible trajectory, stable against fluctuations, and computable from the pipeline. The spectral admissibility programme produces, for each prime q and each conjugate pair $(c, q-c)$, a family of traces $\sigma_{\text{pair}}(n)$ indexed by BFS depth n [**?**]. We show that the correct variable is not $\sigma_{\text{pair}}(n)$ itself, but its complement $I(n) = \sigma_{\text{pair}}(0) - \sigma_{\text{pair}}(n)$.

2 The Two Variables and Why σ_{pair} Is Not the Right One

The observable $\sigma_{\text{pair}}(n) = \sigma_c(n)\,\sigma_{q-c}(n)$ measures the residual projective capacity of the conjugate pair at depth n: it counts, normalised by q, the fraction of Weil fingerprint vectors that remain linearly independent of the Gram–Schmidt span built over the preceding shells. By construction, $\sigma_{\text{pair}}(n)$ decays as n grows and cannot serve as a temporal proxy, since a decreasing quantity does not define an ordering compatible with the accumulation of structure. The correct variable is its complement.

Definition 1 (Cumulative projected capacity)**.** *Let $\sigma_{\text{pair}}(n)$ denote the canonical pair observable [?] at BFS depth n. The* cumulative projected capacity *at depth n is*

$$I(n) \;:=\; \sigma_{\text{pair}}(0) - \sigma_{\text{pair}}(n). \tag{1}$$

$I(n)$ measures the spectral structure that has already been stably projected up to depth n: it starts at zero and accumulates as the cascade progresses, with $I(0) = 0$ and $I(n) \to \sigma_{\text{pair}}(0)$ as $\sigma_{\text{pair}}(n) \to 0$.

Remark 1. *$I(n)$ is non-decreasing if and only if $\sigma_{\text{pair}}(n)$ is non-increasing. While this is expected under a dissipative interpretation of the BFS cascade, the present work establishes it empirically without relying on a formal dissipation theorem: the monotonicity of $I(n)$ is observed directly, step by step and pair by pair, across all tested primes.*

3 Temporal Ordering Conjecture

Proposition 1 (Projective temporal ordering — numerical form)**.** *Define an ordering on effective states along the BFS cascade by*

$$U_t \prec U_{t+1} \quad\Longleftrightarrow\quad I(U_t) \le I(U_{t+1}). \tag{2}$$

For $q \in \{29, 61, 101, 151\}$, this ordering is strictly observed along the tested BFS trajectories: $I(n)$ is strictly non-decreasing at every step, for every conjugate pair $(c, q-c)$, with no physically meaningful exception.

Remark 2 (Status)**.** *This is numerical evidence, not a proof. The result establishes the ordering empirically along the tested trajectories and does not exclude the possibility of exceptions at larger q or along non-BFS paths. The one-way activation property—that modes, once stably projected, do not deactivate—remains to be established analytically.*

4 Protocol and Data

Computations use the `o25_paired_pipeline.py` pipeline with parameters `M=50` independent block samples per pair and `bfs_frac=0.99` (full graph exploration). Monotonicity of $I(n)$ is tested by the `o_stability_diagnostic.py` script, which verifies, for each prime and each pair, that no step $I(n{+}1) < I(n)$ occurs after the burn-in depth $n_{\text{burn}} = 2$.

Table 1: Monotonicity and stability of $I(n)$ per prime. "Steps" counts BFS depths after burn-in. "Mean mono" reports whether the pair-averaged $I(n)$ is strictly non-decreasing. "Regressions" counts pairs with at least one physically meaningful decrease in I (threshold: relative decrease $> 10^{-6}$ of the plateau value).

q	pairs	steps	mean mono	regressions	mono frac
29	14	25	YES	0	1.0000
61	30	55	YES	0	1.0000
101	50	92	YES	0	1.0000
151	75	138	YES	0*	1.0000*

*At $q = 151$, 70 pairs exhibit floating-point fluctuations of order 10^{-10}–10^{-13} on the saturation plateau, where $I/I_{\max} = 1.000000$. These are $\sim$6 orders of magnitude below the mean positive step size of 4×10^{-2} and are not counted as physical regressions.

Figure 1: Left: residual capacity $\sigma_{\text{pair}}(n)$ (strictly decreasing) for $q = 101$, all 50 pairs shown. Right: cumulative projected capacity $I(n) = \sigma_{\text{pair}}(0) - \sigma_{\text{pair}}(n)$ (strictly non-decreasing) for the same pairs. The two panels are exact complements and together visualize the one-way activation property. The plateau in the right panel corresponds to Born–Infeld saturation at the pair level.

5 Interpretation

One-way activation. The zero-regression result means the admissible BFS cascade behaves as a one-way activation process in spectral space. Once a mode contributes stably to the Gram–Schmidt span, it does not leave: the spectral structure accumulated at depth n is preserved at all subsequent depths. This is the empirical content of the stability condition required for the identification of $I(n)$ as a temporal proxy: an ordering defined by I is well-posed only if I never reverses.

Saturation plateau and Born–Infeld bound. $I(n)$ reaches its maximum $\sigma_{\text{pair}}(0)$ when the residual capacity is exhausted. This plateau is interpreted as Born–Infeld saturation at the pair level: once the projective flux budget of the conjugate pair is fully consumed, no additional spectral structure can be resolved within the effective description. Saturation is irreversible in all tested cases; this is consistent with the bounded-flux constraint of the Cosmochrony framework [1] but is not derived from it here.

Candidate arrow of time. The strict monotonicity of $I(n)$ provides a candidate ordering compatible with an arrow of time, without invoking thermodynamic entropy or a background temporal coordinate. The ordering is projective: it reflects the irreversible accumulation of stably activated spectral structure along the admissible cascade. Whether this ordering coincides with the physical time parameter of the full framework [1] is left as an open question; the present result establishes only its monotone and stable character along the tested BFS trajectories for $q \in \{29, 61, 101, 151\}$.

6 Open Questions

1. Does strict monotonicity of $I(n)$ hold for all admissible primes q, or are there exceptions at large q? Extension to $q \in \{211, 307, 401\}$ is pending; preliminary data at $q = 151$ are consistent with the property (Table 1).

2. Can the one-way activation property—$\sigma_{\text{pair}}(n)$ non-increasing along the full cascade—be proved analytically from the structure of the BFS expansion and the admissibility constraints of χ?

3. What is the precise relationship between $I(n)$ and the capacity exponent δ_{pair} [**?**]? Is $I(n)$ sensitive to the value of δ_{pair} or only to its sign?

4. Does the saturation depth (the n at which $I(n)$ plateaus) carry physical meaning independent of q?

5. What is the precise relationship between BFS shell depth and effective resolution-cell entry? Preliminary numerical evidence suggests that the characteristic shell n^* at which $\sigma_{\text{pair}}(n)$ first undergoes a qualitative knee increases with q over the tested cases $q \in \{29, 61, 101\}$, with approximate values $n^*(29) \approx 4$, $n^*(61) \approx 6$, $n^*(101) \approx 8$. This suggests that BFS shells do not define resolution cells, but may reveal the depth at which such cells become effective. The structural status, scaling law, and analytical derivation of this relation are left open.

7 Conclusion

The observable $\sigma_{\text{pair}}(n)$ measures residual projective capacity and decays along the BFS cascade; its complement $I(n) = \sigma_{\text{pair}}(0) - \sigma_{\text{pair}}(n)$ measures cumulative projected capacity and grows. Across $q \in \{29, 61, 101, 151\}$, covering 169 pairs and 310 post-burn-in steps, $I(n)$ is strictly non-decreasing at every BFS step and for every conjugate pair, with no physically meaningful regression. This establishes $I(n)$ as a robust monotone proxy for temporal ordering along the admissible spectral cascade, consistent with the identification of time with the irreversible accumulation of stably projected structure in the Cosmochrony framework [1]. The one-way activation property observed here is a numerical result; its analytical derivation from the admissibility constraints of Π and the scaling of $n_1(q)/q$ [**?**] remains the central open direction.

References

[1] Jérôme Beau. Relational reconstruction of spacetime geometry from graph laplacians, 2026. Cosmochrony companion paper A.